

MACHINE LEARNING IN PHYSICS TUTORIAL 1

HARRISON B. PROSPER

PHY6937 / PHY4636

Installation of miniconda3

1. Go to website:
<https://www.anaconda.com/docs/getting-started/miniconda/install>
2. Download the installer for your operating system and follow the prompts to install **miniconda3**. Then...
3. Create a miniconda3 environment (e.g., here named **mlp**)

```
conda create -n mlp
```

(use `conda env remove -n name` to remove the environment “name”).)
4. Activate environment

```
conda activate mlp
```
5. Install PyTorch:

```
conda install pytorch
```

For
PyTorch,
check for

pytorch
numpy
sympy

The following packages will be downloaded:

package	build	
blas-1.0	openblas	10 KB
deprecated-1.2.13	py313hca03da5_0	18 KB
filelock-3.17.0	py313hca03da5_0	39 KB
fsspec-2025.7.0	py313h7eb115d_0	655 KB
gmp-6.3.0	h313beb8_0	494 KB
gmpy2-2.2.1	py313h5c1b81f_0	216 KB
importlib-metadata-8.5.0	py313hca03da5_0	54 KB
libabseil-20250127.0	cxx17_h313beb8_0	1.2 MB
libgfortran-5.0.0	11_3_0_hca03da5_28	142 KB
libgfortran5-11.3.0	h009349e_28	1.0 MB
libopenblas-0.3.29	hea593b9_0	10.1 MB
libprotobuf-5.29.3	h14f15fd_1	2.8 MB
libtorch-2.6.0	cpu_openblas_h5ebe3f5_6	29.9 MB
libuv-1.48.0	h80987f9_0	432 KB
llvm-openmp-19.1.7	h3b2fb71_2	325 KB
mpc-1.3.1	h80987f9_0	119 KB
mpfr-4.2.1	h80987f9_0	456 KB
mpmath-1.3.0	py313hca03da5_0	1000 KB
networkx-3.4.2	py313hca03da5_0	3.1 MB
nomkl-3.0	0	10 KB
numpy-2.3.1	py313h50dd0cd_0	13 KB
numpy-base-2.3.1	py313h2506b34_0	6.7 MB
opentelemetry-api-1.30.0	py313hca03da5_0	99 KB
pytorch-2.6.0	cpu_openblas_py313h9475015_6	30.5 MB
setuptools-72.1.0	py313hca03da5_0	2.6 MB
sleef-3.5.1	h80987f9_2	357 KB
sympy-1.13.3	py313hca03da5_1	15.0 MB
wrapt-1.17.0	py313h80987f9_0	64 KB
zipp-3.21.0	py313hca03da5_0	31 KB
Total:		107.3 MB

Installation of miniconda3

6. Then install packages:

```
conda install jupyterlab  
conda install matplotlib  
conda install pandas  
conda install scipy  
conda install scikit-learn  
conda install ffmpeg  
conda install tqdm
```

Installation of **mlinphysics**

7. Create a folder for your work this semester (on Linux, macOS)

```
mkdir mlprojects  
cd mlprojects
```

8. Download from GitHub the package **mlinphysics**

```
git clone https://github.com/hbprosper/mlinphysics
```

From your current folder (e.g., mlprojects) do

```
cd mlinphysics  
pip install -e .
```

9. Navigate to **mlprojects** and launch **jupyterlab**,

```
jupyter lab
```

which should appear in your default browser.

Installation of `mlinphysics`

Please NOTE:

1. Work in your own folder (e.g., `mlprojects`), NOT in `mlinphysics`!
2. The package `mlinphysics` may be improved, so code etc. could change and require a periodic `git pull` to update it.

Tutorial 1

1. Copy the following notebooks from **mlinphysics** to your working folder
 1. `test.ipynb`
 2. `python_minimum_part1.ipynb`
 3. `python_minimum_part2.ipynb`
2. In your jupyterlab environment:
 1. Test your installation by executing the `test.ipynb` notebook.
 2. Work through the python minimum tutorials *line-by-line* and try to understand what each line or set of lines does.